

LSC PUBLIC DATA & CONSTRAINT RECONNAISSANCE

Final Pre-BEST-2 Freeze Readiness Assessment

Executive Summary

This report constitutes a final, unprotective, and rigorous data reconnaissance for the LSC (Liquid Scintillator Core) framework prior to the BEST-2 experimental phase. Operating under the strict assumption that LSC is an unvalidated phenomenological framework, this audit maps all publicly available datasets, evaluates their sufficiency for likelihood-level analysis, identifies critical missing metadata, and maps external physics constraints (solar, KATRIN, IceCube).

The central finding is that while the public domain offers sufficient data to establish the absolute deficit scale (trace), the framework remains fundamentally underconstrained regarding its directional and time-dependent parameters (traceless tensor components) due to the proprietary nature of extraction-level covariance and sidereal timestamps.

A. Map of Publicly Available Datasets and Resources

Resource	Source / Link	Contents & Machine Readability	Sufficient?	Class
BEST (2022)	PRL 128, 232501 (2022) arXiv:2112.14599	Inner/Outer zone rates (20 extractions). <i>Machine-readable:</i> No (PDF tables).	No. Lacks full covariance matrix.	ESSENTIAL
GALLEX	PRD 80, 082001 (2009)	Historical Cr-51 source calibration rates. <i>Machine-readable:</i> No.	No. No timestamp event logs.	IMPORTANT
SAGE	PRC 80, 015807 (2009)	Cr-51 and Ar-37 calibration extractions. <i>Machine-readable:</i> No.	No. Summary statistics only.	IMPORTANT
KATRIN	Nature Phys 18, 160 (2022)	Sub-eV neutrino mass limits. <i>Machine-readable:</i> Yes (Data releases).	Yes. Bounds eV-scale steriles.	IMPORTANT
IceCube	Nature Phys 14, 961 (2018)	Lorentz Violation / SME tensor limits. <i>Machine-readable:</i> Yes (Data releases).	Yes. Constrains high-energy anisotropy.	OPTIONAL
Solar Neutrinos	Borexino: Nature 562 (2018)	pp, Be-7, B-8 flux survival probabilities. <i>Machine-readable:</i> Yes (Borexino portal).	Yes. Bounds total disappearance.	ESSENTIAL
Cross-Section	PLB 795, 542 (2019)	Gallium cross-section uncertainties. <i>Machine-readable:</i> No.	No. Theory limits remain debated.	ESSENTIAL

Resource	Source / Link	Contents & Machine Readability	Sufficient?	Class
Covariance	N/A (Baksan collaboration)	Correlated systematics across BEST zones. <i>Machine-readable:</i> N/A.	N/A.	UNAVAILABLE

B. Missing Information & Validation Blockers

Fundamental Underconstraint: The LSC framework introduces a detector-frame tensor (D_{ij}) allowing for directional and orientation-dependent variation. The public data is **insufficient** to constrain this.

- **Missing Covariance:** The lack of a public, fully correlated systematic covariance matrix for the BEST inner/outer zones prevents rigorous χ^2 testing. Treating the zones as having diagonal independent errors artificially inflates the statistical significance of any fit.
- **Missing Timestamps & Sidereal Data:** To fit the traceless (directional) parameters of the LSC framework, precise time-stamped event data for extraction runs must be mapped to celestial coordinates (ICRS). This metadata is completely absent from public archives.
- **Private Access Requirement:** Public data alone is entirely insufficient for full likelihood-level validation. Falsifying or proving the orientation-dependent parameters of LSC requires direct collaboration access to proprietary Baksan hardware logs and extraction chronologies.

C. BEST-2 Resolution Capability

Can BEST-2 realistically resolve the ambiguity?

Partially. BEST-2 aims to refine measurements, potentially altering target geometries or baseline distances. If BEST-2 identifies a clear oscillation-like wave (distance-dependent), it strongly supports the standard sterile-neutrino hypothesis. If the deficit remains uniformly isotropic across new geometries, it supports either a normalization error (cross-section) or the isotropic trace parameter of LSC.

However, **detector/systematics degeneracy will remain** unless BEST-2 specifically publishes continuous, time-resolved extraction data mapped to celestial coordinates. Without this, an LSC tensor fit is indistinguishable from localized chemical inefficiencies or transient noise.

D. Requisite Packages & Constraint Maps

1. Minimum Required Dataset Package

- Official extraction rates for BEST, SAGE, and GALLEX (Publicly available).
- Gallium cross-section theoretical bounds (Bahcall/Kostensalo) (Publicly available).
- Exact spatial coordinates of all target chamber zones relative to the Earth's rotational axis (Private/Limited).

2. Critical Missing Information List (The Blockers)

- Run-by-run fully correlated covariance matrices for BEST.
- Absolute UTC timestamps for the initiation and conclusion of each chemical extraction.
- Counter-level efficiencies and dead-time hardware logs.

3. External Constraint Map

- **Solar/Atmospheric Consistency:** Limits maximum isotropic disappearance. Constrains LSC's ϕ_D to avoid violating Borexino limits.
- **KATRIN:** Prevents the framework from assuming heavy sterile masses beyond ~ 0.8 eV as a baseline comparator.
- **Standard-Model Extension (SME) Limits:** IceCube searches place stringent limits on Lorentz-violating tensor fields. If LSC assumes a cosmological background tensor, it must conform to IceCube's SME bounds ($< 10^{-24}$).

4. Pre-BEST-2 Freeze Readiness Assessment

The framework is mathematically mature enough to be frozen, but it is **experimentally unready** for true falsification due to the public data vacuum. It must be frozen as a "conditional template" awaiting private data, rather than an immediately testable hypothesis.

E. Final Decision

RECOMMENDATION: FREEZE WITH DOWNGRADED CLAIMS

Based on the severe lack of public covariance and orientation metadata, the LSC framework should be **FROZEN** in its current mathematical state (v6.3.0), but its operational status must be strictly limited.

It cannot be presented as a fully verifiable theory using only public data. The freeze must explicitly state that the tensor (directional) components of LSC are currently untestable and that the framework serves solely as a preregistered mathematical pipeline, completely dependent on future collaboration with the BEST-2 team for sidereal and hardware metadata.